

# Saee Desai

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## SUMMARY

**Applied AI and Machine Learning Engineer** with experience building NLP, multimodal, and AI inference systems using transformers, diffusion models, and production-style ML pipelines. Skilled in model fine-tuning, evaluation, and scalable workflow design across moderation, generative AI, and secure data applications.

## EDUCATION

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| <b>University of Maryland, College Park</b><br><i>M.S. in Data Science, GPA: 3.85/4</i><br><b>Vishwakarma Institute of Information Technology</b><br><i>B.Tech in Computer Engineering, CGPA: 9.13/10</i> | Maryland, USA<br><i>Aug 2024 – May 2026</i><br>Pune, India<br><i>Aug 2020 – May 2024</i> |
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## TECHNICAL SKILLS

**Languages:** Python, SQL, C++  
**Machine Learning & AI:** PyTorch, TensorFlow, scikit-learn, Transformers, CNNs, Diffusion Models, LoRA, CLIP  
**NLP & Generative AI:** HuggingFace, LangChain, LangGraph, Text Classification, Emotion Analysis  
**Data & Evaluation:** Pandas, NumPy, Feature Engineering, Model Evaluation, Error Analysis, Statistical Analysis  
**Systems & Tools:** Git, Linux, Flask, REST APIs, AI Inference Pipelines, Gradio, Jupyter, AWS S3 (basic)

## PROJECTS

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                     |
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| <b>Real-Time Reddit Moderation and Emotion Analysis System</b><br>– Built a real-time NLP moderation pipeline using fine-tuned <b>DistilBERT</b> models for hate speech detection and emotion analysis across Reddit posts and nested comment threads.<br>– Fine-tuned transformer models on <b>HateXplain</b> and <b>GoEmotions</b> datasets, achieving <b>76.25% hate detection accuracy</b> while optimizing for low-latency inference.<br>– Designed a modular <b>Flask-based inference pipeline</b> with model caching, confidence scoring, and thread-aware analysis for scalable moderation workflows.<br>– Developed human-in-the-loop moderation features for interpretable prediction review and deployment-oriented evaluation.                                                           | Aug 2025 – Dec 2025 |
| <b>Emotion-to-Art Generation using Transformers and Diffusion Models</b><br>– Designed an <b>end-to-end multimodal system</b> that converted emotion text into artwork using transformer-based emotion embeddings and diffusion models.<br>– Trained a <b>DistilBERT emotion classifier</b> across <b>27 classes</b> and mapped probabilistic emotion outputs to image generation controls for improved alignment between user intent and generated output.<br>– Fine-tuned <b>Stable Diffusion with LoRA</b> under constrained GPU memory by optimizing precision, batch size, and training configuration to stabilize convergence.<br>– Built an evaluation pipeline using <b>CLIP similarity</b> and qualitative feedback analysis to identify systematic generation failures in subtle emotions. | Aug 2025 – Dec 2025 |
| <b>Post-Quantum Secure Pharmacovigilance Pipeline</b><br>– Designed a secure healthcare data pipeline integrating <b>ML-KEM-768</b> , <b>ML-DSA-65</b> , AES-256-GCM, and HKDF for post-quantum secure data exchange and verification.<br>– Built modular hospital, gateway, attacker, and pharmaceutical receiver systems with <b>Flask APIs</b> , cryptographic validation workflows, and packet-level logging.<br>– Simulated replay, corruption, and ciphertext tampering attacks while benchmarking cryptographic stages across multiple healthcare file formats up to <b>10 MB</b> .                                                                                                                                                                                                           | Jan 2026 – May 2026 |
| <b>Facial Expression Recognition with Deep Learning</b><br>– Built CNN and transfer learning models ( <b>MobileNetV2</b> , <b>ResNet</b> , <b>VGG</b> ) to classify facial expressions on a <b>39,000+ image dataset</b> .<br>– Mitigated severe class imbalance using targeted augmentation and statistical analysis, improving minority-class recall and achieving <b>80%</b> test accuracy.<br>– Performed per-class error analysis and cross-validation to identify recurring failures in subtle emotions and guide model refinement.                                                                                                                                                                                                                                                            | Aug 2024 – Dec 2024 |

## PUBLICATIONS

- **Enhanced Communication Security: Leveraging Quantum Cryptography.**  
*ICT Analysis and Applications (ICT4SD 2024)*, Lecture Notes in Networks and Systems, vol. 1197. Springer, March 2025.